

Name:

Oisín Argand

$$\frac{130}{130} = 100\%$$

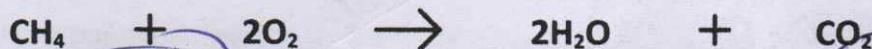
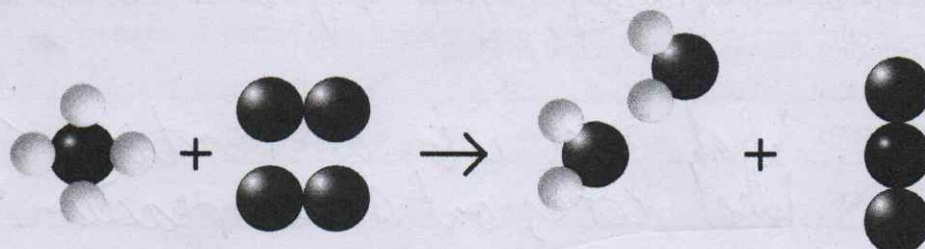
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Question 1

Question 6

May 26 (15 marks)

- (a) Explain, using the diagram below, the Law of Conservation of Mass.



number

+ 3

3

The particles on the left hand side stay the same, even when mixed up on the right hand side. This demonstrates how in a reaction, nothing is created or destroyed, just rearranged.

- (b) Is the above reaction an example of a physical or a chemical change? Verify your answer.

It is a chemical change because new substances are being made.

- (c) State whether the following are examples of a physical or a chemical change by putting a tick (✓) in the correct column.

	Physical	Chemical
Ice cube melting when taken out of a freezer	✓	
Burning a candle in a room		✓
Rust forming on a bicycle chain		✓

3

Question 2

Question 15

(45 marks)

Some chemical reactions proceed quickly and others proceed at a slower rate.

- (a) Describe what a *chemical reaction* is.

A reaction in which the reactants are altered chemically, forming a new substance

- (b) Distinguish between *reactants* and *products*.

Reactants: What goes into a reaction ✓
Products: What comes out of a reaction ✓

- (c) Study the chemical reaction given below and name the *products* and the *reactants* in this reaction:



Products: Magnesium chloride + Hydrogen
Reactants: Magnesium + Hydrochloric acid ✓

Question 3

- (i) Describe what is meant by a *biochemical reaction*.

A biochemical reaction is a chemical reaction that takes place in a biological organism

- (ii) Give an example of a biochemical reaction.

Enzymes in your saliva helping to digest food

(15 marks)

The passage below is about the three states of matter.
The following five words are missing from the passage:

chemical conservation evaporation melting physical

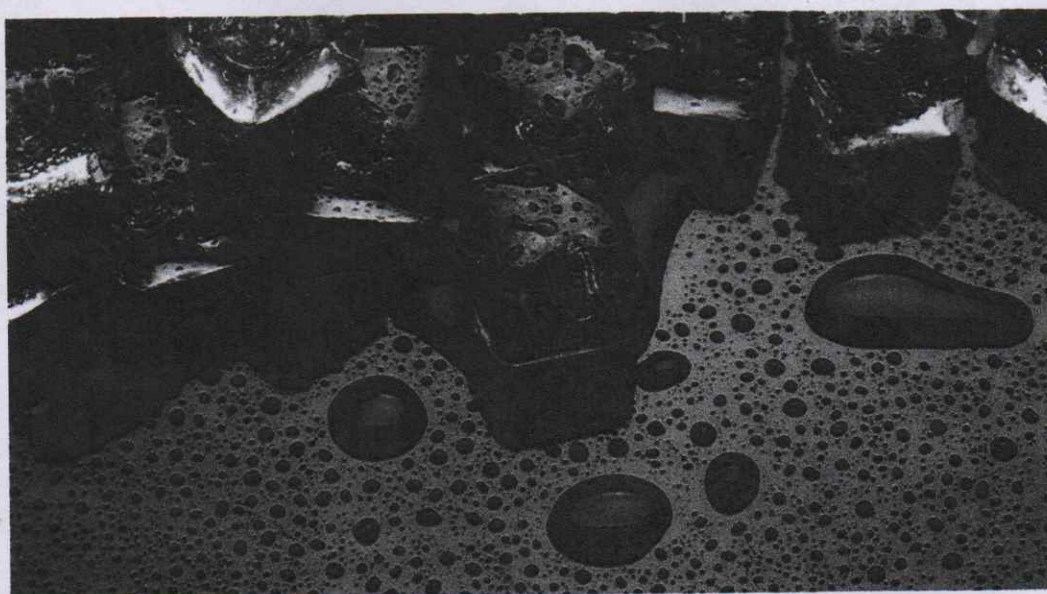
Write the missing words in the spaces provided.

There are three states of matter: solid, liquid and gas. When a solid is heated it turns into a liquid – this change of state is called melting. When a liquid is heated it turns into a gas – this change of state is called evaporation.

Liquid water freezes to become solid ice; this is an example of a physical change. However, when electricity is passed through liquid water it is converted into its elements, hydrogen and oxygen; this is an example of a chemical change.

When liquid water freezes, the mass of the ice formed is the same as the mass of the liquid water. This is an example of conservation of mass.

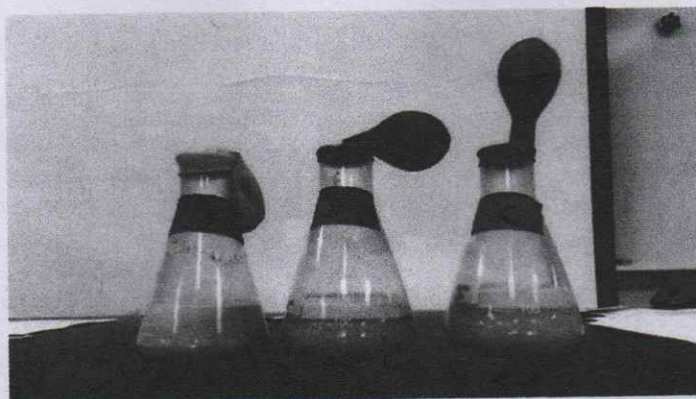
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Question 5

Micro-organisms such as yeast also release gases.

The photograph below shows an investigation into the factors that affect the volume of gas released by yeast.



A (5 °C) B (20 °C) C (35 °C)

In each case, the gas released is collected in the balloon attached to the top of the reaction flask.

Test **A** was carried out at 5 °C.

Test **B** was carried out at 20 °C.

Test **C** was carried out at 35 °C.

As the reaction temperature increased (from **A** to **C**), a greater volume of gas was collected in the balloon.

- (g) Suggest two reasons why an increase in reaction temperature might lead to an increase in the volume of gas collected.

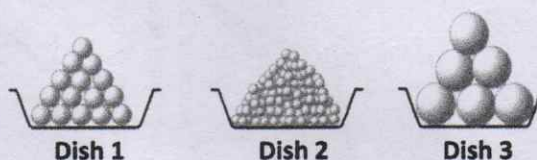
Higher temperatures speed up particles, which allow for ^{more} reactions to take place ~~which~~. Higher temperatures also result in better activation of yeast.

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Question 6

The rate of chemical reactions is influenced by surface area.

The diagram shows three dishes, 1, 2 and 3. Each dish contains 2 g of marble chips.



- (c) (i) Name an instrument that could be used to measure 2 g of marble chips.

Electric balance

- (ii) Which dish contains marble chips with the largest total surface area? Put a tick (✓) in the correct box.

Dish 1 ☐

Dish 2 ☒

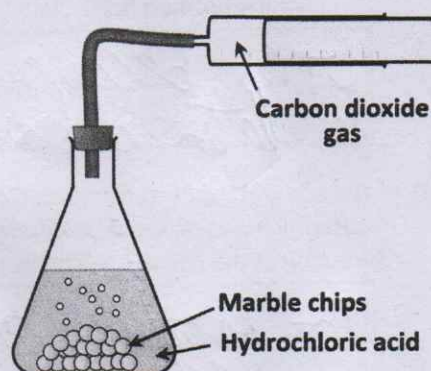
Dish 3 ☐

Marble chips react with hydrochloric acid to produce carbon dioxide gas.

A group of students was asked to investigate the effect of marble chip size on the rate of this reaction.

They used the apparatus shown on the right.

Some of the variables that might change or be kept constant during this investigation are listed below.



The concentration of hydrochloric acid	The size of the marble chips
The volume of gas produced each minute	The loudness of the noise in the room
The temperature of the acid	The total mass of the marble chips

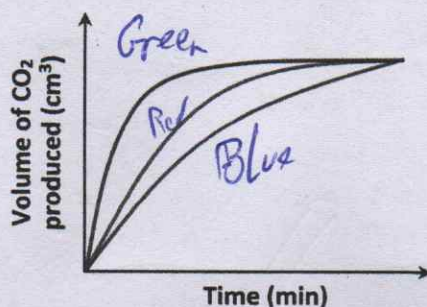
- (d) (i) From the list, identify three variables which the students should have kept constant during the investigation.

- The temperature of the acid
- The concentration of hydrochloric acid
- The total mass of the marble chips

- (ii) From the list, identify one relevant variable which changed during the investigation.

The size of the marble chips

- (e) The graph below shows the results obtained by the students.



- (i) Which of the curves (green or red or blue) on the graph corresponds to the reaction with the slowest rate? Justify your answer.

Blue because it took the longest time to reach the same volume of CO_2 produced as the other two (6)

- (ii) Which of the curves (green or red or blue) on the graph corresponds to the reaction which finished first? Justify your answer.

Green because it took the shortest time to flatten out at the final volume of CO_2 produced (6)

- (f) The students collected carbon dioxide gas in this investigation.

Describe how you would produce another common gas in the laboratory, including how you would test the gas to confirm its identity.

To produce oxygen in the laboratory, you set up a Buchner flask with tubing through a basin of water, up a beehive shelf and into an upside down graduated cylinder. You put hydrogen peroxide and manganese dioxide into the Buchner flask and seal it. Bubbles of oxygen will appear in the graduated cylinder. To test for oxygen, light a splint then blow it out, leaving it glowing. If put into oxygen, the splint will relight.

Excellent.

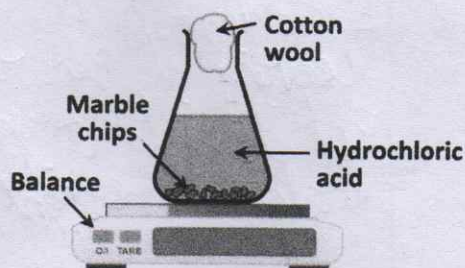
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Question 7

The students carried out a second investigation.

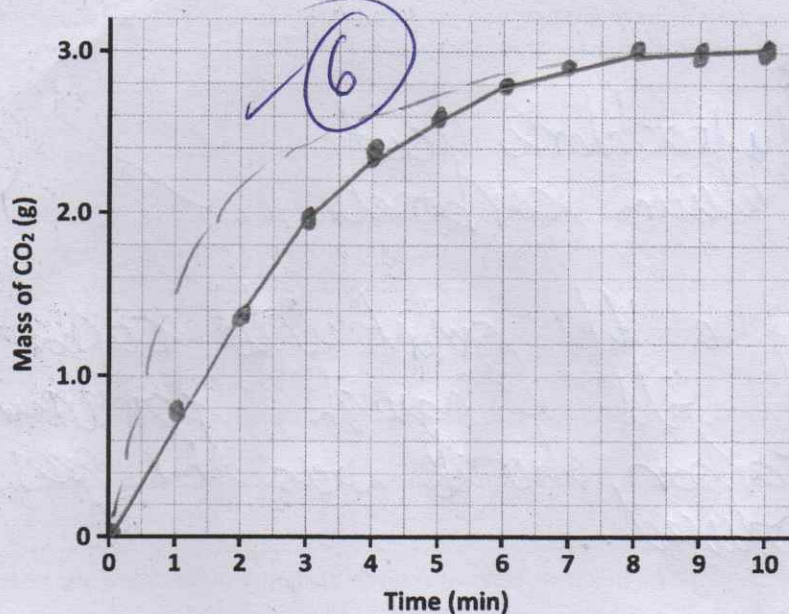
This time they measured the mass of carbon dioxide produced at regular intervals using the apparatus on the right.

The data collected by the students are shown below.



Time (min)	0	1	2	3	4	5	6	7	8	9	10
Mass of CO ₂ (g)	0	0.8	1.4	2.0	2.4	2.6	2.8	2.9	3.0	3.0	3.0

(d) In the space below, draw a line graph of the students' results.



(e) At what time was the reaction complete?

8 minutes (3)

(f) On your graph above, draw a dashed line (-----) to show the result you would expect if the above investigation was repeated exactly as before, but using smaller marble chips of the same total mass.

Question 8

Question 12

(30 marks)

Photosynthesis is described as plants making their own food using light.

- (b) One of the products of photosynthesis is O_2 . Describe a test for the presence of this gas.
Draw a labelled diagram to support your description.

Method:	Diagram
Extinguish a lit splint, leaving it glowing. Put it into oxygen and watch it relight	

Question 9

- (e) As part of your Science studies, you prepared carbon dioxide in the laboratory. Name the two chemicals you used to make this gas.

1. Hydrochloric acid
2. Calcium carbonate

- (f) Describe any one test used to confirm the gas produced was carbon dioxide.

Put a lit splint into carbon dioxide. It will extinguish, confirming carbon dioxide was the gas produced.

- (g) There is frequently a negative association with carbon dioxide. State any one positive use of this gas.

Used to extinguish fires

Question 10

- (f) Provide an explanation why chopped potatoes cook faster than whole potatoes.

They have a larger surface area, and thus react faster.